BSc 2023/2024

For internal students of Imperial College.

Mathematics Foundations (BUSI40001)

2 hours

Sample Exam, Dec 2023

Instructions for the exam

- For Section A, internet access with ChatGPT is allowed and a computer with R or Python is needed.
- Section B is closed-book and paper-based. Section B is to made accessible to the students after completing Section A.
- Answer ALL questions (Please note that not all questions carry equal marks, the mark breakdown is indicated throughout the exam).
- It is important to write neatly and clearly.
- If a question or part of a question is unclear to you, state any assumptions you need in order to complete the question.

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SECTION A

You have 45 minutes for Section A. Answer all questions. You must close your laptop after 45 minutes.

1. (23 marks) The XL file

`SampleExamData_2023.xlsx`

contains data for annual returns on the S&P 500 Index and Berkshire Hathaway.

This question is about using OLS linear regression to analyze the relationship between returns for Berkshire Hathaway and returns on the S&P 500 Index.

(a) Write and use Python or R code to read the data from the XL file and create a column vector $\underline{x}$ containing returns for the S&P 500 Index and a column vector of the same size, $\underline{y}$, containing returns for Berkshire Hathaway. [4 marks]

(b) Write out the linear regression of Berkshire Hathaway returns against S & P 500 returns in the matrix form

$$\underline{y} = \mathbf{X}\underline{\beta} + \underline{\epsilon},$$

where $\mathbf{X}$, $\underline{\beta}$, and $\underline{\epsilon}$ are to be defined. [2 marks]

(c) Show that the expression for $\underline{\beta}$ which minimizes $\underline{\epsilon}^\top \underline{\epsilon}$ is given by

$$\underline{\widehat{\beta}} = \left(\mathbf{X}^T\mathbf{X}\right)^{-1}\mathbf{X}^T\underline{y}.$$

[5 marks]

(d) Show that the fitted values are given by

$$\underline{\hat{y}} = \mathbf{H}\underline{y}$$

where

$$\mathbf{H} \equiv \mathbf{X}\left(\mathbf{X}^T\mathbf{X}\right)^{-1}\mathbf{X}^T$$

[3 marks]

(e) Show that $\mathbf{H}$ is idempotent and symmetric. [4 marks]

(f) Write and use R or Python code to find $\underline{\widehat{\beta}}$. [5 marks]

2. (a) Find and classify all critical points (local maximum, local minimum or saddle point) of

$$(x+y)^2 - 3xy^2.$$

You may write code in R or Python to carry out your calculations and draw any plots. [12 marks]

(b) Find the local maxima and minima of $(x+y)^2 - 4y$ on the circle $x^2 + y^2 - 16 = 0$. You may use Python or R to perform calculations. [8 marks]

SECTION B

You have 1 hour 15 minutes for Section B. Answer all questions.

1. (34 marks) Consider a real sequence $(a_n)_n$

(a) What does it mean to say that a sequence of real numbers $(a_n)_n$ converges to $\ell \in \mathbb{R}$? [2 marks]

(b) What does it mean to say that a sequence of real numbers $(a_n)_n$ is bounded? [1 mark]

(c) Prove that every convergent sequence is bounded. [6 marks]

(d) Show that if $0 < c < 1$, then the sequence $(c^n)_n$ converges to 0. [4 marks]

(e) Let $(a_n)_n, (x_n)_n, (b_n)_n$ be sequences of real numbers. Suppose that for all $n \in \mathbb{N}$, $a_n \leqslant x_n \leqslant b_n$ and $(a_n)_n, (b_n)_n$ converge to $L \in \mathbb{R}$. Prove that $(x_n)_n$ converges to L too. [5 marks]

(f) Let $(a_n)_n$ be a sequence of positive real numbers. For the following statements provide either a proof or a counterexample.

i. If $\frac{a_{n+1}}{a_n} \to 1$ then (a_n) converges. [6 marks]

ii. If $(a_n)_n$ converges then $\frac{a_{n+1}}{a_n} \to 1$. [10 marks]

2. (33 marks)

(a) Find the orthogonal projection of the vector $\underline{b} = (1, 2, 6)^\top$ onto the plane $x + y + z = 0$ in $\mathbb{R}^3$, where the inner product is the standard scalar product. [4 marks]

(b) Let V be the vector space of real polynomials of degree $\leqslant 2$ and define the inner product as follows

$$\begin{aligned} \langle \cdot, \cdot \rangle : V \times V &\longrightarrow \mathbb{R} \\ (f, g) &\longmapsto \frac{1}{3} \sum_{k \in \{-1,0,1\}} f(k) \cdot g(k) \end{aligned}$$

i. Let U be the subspace of V spanned by 1 and x.

A. Show that 1 and x are orthogonal and use this to find an orthonormal basis for U. [5 marks]

B. Find a set which serves as an orthonormal basis for V and then prove it is indeed a basis of V. [10 marks]

(c) Consider a finite dimensional vector space V over a field $\mathbb{F}$ with subspaces U and W.

i. What is meant by the sum $U + W$ of U and W ? [2 marks]

ii. Show that $U + W = \operatorname{span}(U \cup W)$. When is $U + W = U \cup W$? Justify your answer. [12 marks]

End of Paper